

How much drinkable water does AI actually use?

Posted by u/Taramund • 0 points • 10 comments

I think we all heard about how AI companies use absurd amounts of water. My first question - is it usable, potable water that they're wasting?

My general approach to AI is that 1) it isn't going away, at least any time soon, 2) I despise using it for art and misinformation (videos, images, online bots, books, etc.), 3) it's quite useful in the things it's good at - summaries, first phase of online research, etc.

My hope is that I'm not wasting huge amounts of potable water when I use AI to study (summarise materials, evaluate their usefulness, prepare a study plan, etc.) I feel like generating images (which I usually don't do) must consume more resources than simple prompts, but I don't even know if that's accurate.

I also feel like using it privately in a limited way (like I do) surely can't compare to how much big companies and corporations use it, and so my consumption is surely dwarfed by their usage. That just seems like a flimsy excuse though.

So, to summarise (TLDR): is private, limited use of AI truly harmful to the environment, especially compared to the damage caused by big companies (AI/water and greenhouse gases)? Does it matter whether one only uses text prompts/responses or actually generates images or videos?

Comments

u/LevelMiddle • 2 points • 2026-02-25 16:13 UTC

When i first moved to LA 15 years ago everyone was concerned about water conservation. Then i looked at the numbers. 130 million gallons per year per farm. 30,000 gallons per single person per year at home use. 4000 people's water use (non conservative) equated one average california farm. Seems significant to save water then, right? Hmmm. Then take into account about 90 billion gallons of water are used to water golf courses in california. That'd about 3 million people's annual water usage, about the population of LA.

Yeah if everyone kept their water running while brushing teeth or something, it'd make a big impact. 40

million californians.

But then... data centers use 13 billion in california. Almond farms use more than 1 trillion a year. Total household use by all population is about 1 trillion a year. About 25 trillion gallons of water are used by everything except household use.

This is just in california. In other places with fewer farms and tech i suppose it would change the ratio.

Idk there seems to be significantly more things that use a massively "more" amount of water that we (as individuals) cannot do anything about. Apparently one prompt in chat uses about five drops of water. It's projected to be 1.5 trillion gallons of water used by AI use across the globe over the next few years. We would do a lot better by cutting out almond milk than by not using AI.

u/seriouslyepic • 0 points • 2026-02-25 15:28 UTC

In my opinion, people that yell "AI is using all the water!" are just mad about AI and using the easiest talking point. A better argument or conversation would be that countries/states/cities should be maintaining/upgrading and expanding infrastructure to support the future for both people and technology.

To me, it's similar to someone being mad that too many people are on government welfare while not wanting to acknowledge or discuss why so many people can't afford basic life necessities.

u/bangbangracer • 1 points • 2026-02-25 15:11 UTC

Basins and reservoirs are already stressed enough to maintain human needs and agricultural needs. One US household is estimated to use about 100k to 120k gallons of water per year. That's about 117 trillion gallons per year for the entire country. We can turn that into 322ish billion per day. About 130ish billion is used by power generation, 118ish billion is for agriculture, and the actual use is about 39ish billion. Everything not accounted for is industrial use.

With AI's current demands and plans for expansion, estimates say that 117 trillion gallons per year will nearly double by 2030, and we don't have the reservoirs that can service that or the reclamation abilities to speed up the replenishment of those reservoirs.

u/DiogenesKuon • 2 points • 2026-02-25 15:11 UTC

On the national level that amount of water that is used by data centers is pretty minimal compared to all the other things (it's more a local problem). The total global amount of water used by all data centers, not just AI datacenter, is around 560 billion gallons a year. That's obviously a lot of water, but just in the US we use 760 billion gallons a year to water golf courses.

Further more, at the personal level, there are plenty of things we do that use the equivalent energy as a normal level of AI usage. One ChatGPT query uses about as much energy as a lightbulb running for 30 seconds. While we don't want to leave light bulbs on unnecessarily, we also don't hesitate to use them when they are useful.

u/KronusIV • 1 points • 2026-02-25 15:05 UTC

Most of the "work" an AI does is in the training, not in the answering of questions. So it doesn't make much difference in what you as a consumer do. If you use AI, you're supporting the AI business model, which includes using lots of water for cooling.

How much water can be tricky to say. There are lots of different ways to do the math, depending on

how good or bad you want the numbers to look. A common number is that a data center uses the same amount of water as a town of 10,000-50,000 people.

u/Taramund • 1 points • 2026-02-25 15:36 UTC

>If you use AI, you're supporting the AI business model, which includes using lots of water for cooling.

That's true, though I'm generally not a fan of the "vote with your wallets" idea. Whatever I do, AI is likely here to stay. Companies won't stop training AI, even if most people would stop using it, since it has many possible uses, including in the military. I'm more worried about whether my usage is noticeably or significantly harmful to the environment.

u/KronusIV • 2 points • 2026-02-25 15:58 UTC

It is true that your usage isn't even a blip on the graph. As for the possible uses, that's a tough call. Last I heard no AI company has turned any profit; some are burning billions with no obvious plan as to how they'll actually make money. If the companies involved are all in with no regards for things like customers or profit then your actions matter even less.

u/Lumpy-Notice8945 • 3 points • 2026-02-25 14:58 UTC

Define "huge".

I recomend the youtube video Hank green made about this. It helps ro get some perspective.

And keep in mind its not just AI that is running on big server racks in datacenters, watching a netflix video does use up water in the exact same way as AI does but probably a bit less in terms of per hour usage.

Computers need cooling and air vents arr not enough for big servers, so they need ACs and they evaporate water. In most cases thats just potable tapwater.

But how much they use is difficult to pin down exactly but its not like you use a bathtub per query or anything like that. An LLM uses about as mich electricity as a gaming PC and to cool that you dont even need water, but to cool a room full of them you might need water, so yes a server full of AI that servers thousands of people might use up some water.

u/Taramund • 1 points • 2026-02-25 15:41 UTC

Thanks, I'll definitely watch the Hank Green video.

So you're saying that individual, moderate use of AI is comparable to moderate use of electricity and streaming?

u/Lumpy-Notice8945 • 3 points • 2026-02-25 16:26 UTC

There is way worse things if your concern is water usage, concrete for example. This current trend to blame AI for everything is just totally misleading, there is valid points for critizism, but water usage really isnt the main issue.